

The quantum-mechanical matrix bootstrap: a cribsheet

with the specialisations needed for purely fermionic (matrix SYK) models

Matrix SYK bootstrap project

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Scope. Everything here is drawn from the project library: Lin–Zheng [1, 2], Lin’s TASI lectures [3], Laliberte–McPeak [4], Klebanov et al. [5], Chen [6], Chang et al. [7]. Equation numbers in brackets refer to those papers. Section 5 contains project-specific statements (marked ★) that are derived in `docs/derivations.md` or verified numerically in this repository, not taken from the literature.

1 Objects

State functional.

A linear functional $\phi(O) = \langle O \rangle$ on operators, meant to be $\langle E|O|E \rangle$ for an energy eigenstate (or $\text{Tr } \rho O$ for a stationary ρ , $[\rho, H] = 0$) [TASI (100); LM (1.4)–(1.5)]. Normalisation $\phi(\mathbf{1}) = 1$ (or N). The bootstrap never constructs the state; it constrains ϕ .

Letters and words.

Letters are the elementary operators (X_I, P_I, ψ_α for BFSS; $\Psi_{ij}, \bar{\Psi}_{ij}$ for us). Words are ordered products. At large N the variables are *single-trace* words $\text{tr}(O_1 \cdots O_n)$ with global-symmetry indices contracted into singlets [LZ24 (8); LZ25 eq. after (3)], normalised $\text{tr} = \frac{1}{N} \text{Tr}$, $X \rightarrow X/\sqrt{N}$, so that $\langle \text{tr } O \rangle = O(1)$ [LZ24 (7); LZ25 (3)].

Level (hierarchy).

$\ell(X) = 1$, $\ell(P) = 2$, $\ell(\psi) = \frac{3}{2}$ (for canonical $\{\psi, \psi\} = \delta$ fermions), additive over letters; chosen so that $\ell([H, O]) = \ell(O) + 1$ and $\ell(\{Q, O\}) = \ell(O) + \frac{1}{2}$ [LZ24 §2.5, (25); LZ25 “Hierarchy”; LM §3.2.1]. A “level- L bootstrap” uses positivity matrices built from operators of level $\leq L/2$ (so entries have level $\leq L$) and all equality constraints among variables of level $\leq L$. “Free variables” = variables left after solving the equalities (LZ24 Table 1; LZ25 Table I).

2 The constraint catalogue

Constraint	Statement	Source
Inner-product positivity	$M_{ij} = \langle O_i^\dagger O_j \rangle \succeq 0$ for any finite set $\{O_i\}$: the Gram matrix of $\{O_i E\rangle\}$. Uses only Hilbert-space positivity, so fermions and sign problems are irrelevant. Adding operators can only tighten bounds (principal minors).	LM (1.1)–(1.4); LZ25 (5); TASI (95)–(96)
Stationarity (EOM)	$\langle [H, O] \rangle = 0$ for all O (true in any eigenstate or stationary ρ). For matrix QM take O single-trace; $[H, \text{tr } O]$ is again single-trace.	LM (1.5); LZ25 (4); TASI (100)
Eigenstate condition	$\langle OH \rangle = \langle HO \rangle = E \langle O \rangle$: stronger, resolves individual eigenvalues (“archipelago”), but bilinear in (E, ϕ) hence non-convex; usable by scanning E . Becomes trivial under large- N factorisation.	LM (2.14) & fn. 2; LZ24 (57)–(58), Fig. 4; TASI §6.2
Reality / time reversal	$\langle O_1 \cdots O_n \rangle^* = \langle O_n^\dagger \cdots O_1^\dagger \rangle$; with real wavefunctions $\langle \text{tr } O_1 \cdots O_n \rangle = \pm \langle \text{tr } O_n \cdots O_1 \rangle$ (– for an odd number of P ’s). Halves the variables.	LM (3.34); LZ24 §2.2(2); LZ25 “Constraints”
Algebra / canonical form	(Anti)commutation relations reduce every word to a canonical (e.g. normal-ordered) form; for matrices, cyclicity of tr produces <i>double traces</i> : $\text{tr } O_1 \cdots O_n = \pm \text{tr } O_2 \cdots O_n O_1$ + double trace.	LZ24 (11)–(12), (29); TASI (101)
Large-N factorisation	$\langle \text{tr } A \text{tr } B \rangle \rightarrow \langle \text{tr } A \rangle \langle \text{tr } B \rangle$ (the <i>only</i> place $N = \infty$ enters in the Hamiltonian approach). Makes the constraints quadratic in the variables. Justified by planar dominance; can fail for operators sensitive to flat directions.	LZ24 (46) & §4; TASI §6.1–6.2
Gauge constraint	Singlet state: $\langle \text{tr}(OG) \rangle = 0$ for all O, G the $SU(N)$ generator (e.g. $G = i[X, P] - \Psi\Psi^\dagger - \Psi^\dagger\Psi + 2N$ for one bosonic + one fermionic matrix). Removes an enormous number of variables; without it the Marinari–Parisi SDP was intractable.	LZ24 (13); LZ25 after (1); LM (3.12)–(3.13), §4

Constraint (cont.)	Statement	Source
Global symmetry (Ward identities, blocking)	If the state is G -invariant, $\langle O_R \rangle = 0$ unless R is the singlet, and $\langle \bar{O}_{\bar{R}\bar{r}} O_{Rr} \rangle = \delta_{\bar{R}R} \delta_{\bar{r}r} a_{\bar{R},R}$. Positivity of the big matrix reduces to $a_{\bar{R},R} \succeq 0$ block by block; channels are related by crossing kernels (6j symbols). Conserved charges give $\langle [F, O] \rangle = 0$ and block-diagonalise M by charge.	LZ24 (16)–(24), (32)–(35); LZ25 App. F; LM (3.18) & §3.2.1
Ground-state positivity	$N_{ij} = \langle O_i^\dagger [H, O_j] \rangle \succeq 0$: any perturbation $\sum c_i O_i \Omega\rangle$ has energy $\geq E_0$. Zero-temperature limit of the energy–entropy balance inequality $\langle O^\dagger O \rangle \log \frac{\langle O^\dagger O \rangle}{\langle O O^\dagger \rangle} \leq \beta \langle O^\dagger [H, O] \rangle$. Supplies <i>upper</i> bounds and shrinks peninsulas to islands.	LZ25 (6); TASI (102)–(104); LM (3.20)
Supercharge (SUSY vacuum)	If $Q \Omega\rangle = 0$: $\langle QO \rangle = \langle OQ \rangle = 0$, at large N $\langle \{Q_\alpha, O_\alpha\} \rangle = 0$ for fermionic O . Implies ground-state positivity: $\langle O^\dagger [H, O] \rangle = \frac{1}{8} \langle [Q, O]^\dagger [Q, O] \rangle \geq 0$. The weaker $\langle [Q^\dagger Q, O] \rangle = 0$ was found redundant with EOM.	LZ24 (14), App. A (47)–(54); TASI (128); LM (3.19)
Nonlinear relaxation	Quadratic terms $x_i x_j$ (from factorisation) $\rightarrow$ new variables $\mathcal{Q}_{ij}$ with $\begin{pmatrix} 1 & \vec{x}^T \\ \vec{x} & \mathcal{Q} \end{pmatrix} \succeq 0$ and $M \succeq \mathcal{Q}$ (covariance positivity). Rigorous, convex, weaker than the equality.	LZ25 App. B (B6)–(B7); TASI (98)–(99)

Objective. Minimise (maximise) $\phi(H)$ for a rigorous lower (upper) bound on E_0 (top of spectrum); or fix E and extremise a one-point function to map the allowed region (“peninsula” if unbounded in some direction, “island” if compact). Typical plots: E vs $\langle \text{tr } X^2 \rangle$ (LZ25 Figs. 1–2; LZ24 Figs. 1–2).

3 Mechanics of a run

1. Enumerate variables (single-trace singlet words up to level L ; or, at finite N , all canonical words).
2. Generate and *symbolically solve* the linear equalities (algebra, reality, gauge, Ward, EOM, supercharge) for a set of free variables. This step, not the SDP, dominates cost (LZ24 §3; LM §3.2.2: 8 h of algebra vs 5–10 min per SDPB point).
3. Build the positivity matrices (M , N , relaxation blocks) as linear matrix inequalities in the free variables, one block per (charge, irrep).
4. Solve the SDP. Generic solvers (Mathematica, SCS) suffice at low level; the problems lack strict feasibility, so high level needs arbitrary precision: SDPA-GMP (LZ25 App. A) or SDPB (LM). Bounds should tighten monotonically with level; plateaus followed by jumps happen (LM Fig. 13 and §3.2.2).
5. Sanity checks: (a) feed the exact state (ED or known solution) into every constraint and verify $M \succeq 0$, EOM residual 0; (b) reproduce analytic low-level bounds, e.g. the virial/Polchinski bounds (TASI (114)–(127); LZ25 (7), App. E).

4 Worked micro-examples

(a) **Fermion number and $E \geq 0$ from a 2×2 block** (LM (3.26)–(3.28)): with $\{\Psi, \Psi^\dagger\}$ operators, $\begin{pmatrix} \langle \text{tr } \Psi^\dagger \Psi \rangle & \langle \text{tr } \Psi^\dagger \Psi^\dagger \rangle \\ \langle \text{tr } \Psi \Psi \rangle & \langle \text{tr } \Psi \Psi^\dagger \rangle \end{pmatrix} \succeq$

0 gives $0 \leq F \leq N^2$; combined with the bosonic block it yields $\langle H \rangle \geq 0$ with $E = 0$ only at $F = 0$ for the quadratic superpotential.

(b) **Level-5 bosonic Yang–Mills** (LZ25 App. E): 19 variables, equalities (E4)–(E32) leave 3 free; irrep blocks (E35)–(E38) give the analytic peninsula/island (E39).

(c) **SUSY sum of squares** ($\star$, this project): for $H = \{Q, \bar{Q}\}$ include $Q, \bar{Q}$ among the O_i . Then $\phi(\bar{Q}Q)$ and $\phi(Q\bar{Q})$ are diagonal entries of $M \succeq 0$, so $\phi(H) \geq 0$ is *automatic*; the true content of a lower bound in a lifted sector is whatever forces $\phi(\bar{Q}Q) + \phi(Q\bar{Q}) > 0$.

(d) **Representation-resolved bounds by hand** (KMPT §3.1–3.2): $C_2 \pm \frac{1}{2} A \cdot A = \frac{1}{4} (A \pm A)^2 \geq 0$ and Cauchy–Schwarz with a symmetry-averaged density matrix $\rho_R = \frac{1}{\dim R} \sum_i |e_i\rangle \langle e_i|$ give E -vs-Casimir bounds (3.8), (3.21) that are saturated in Casimir-type fermionic matrix models. This is a two-operator bootstrap in each irrep.

5 Specialisation to purely fermionic matrix models

Setting: p complex fermion matrices $\Psi_{ij}^{(f)}$, $\{\Psi_{ij}, \bar{\Psi}_{kl}\} = \delta_{il}\delta_{jk}$, $\bar{\Psi}_{ij} = \Psi_{ji}^\dagger$, Q cubic in Ψ , $H = \{Q, \bar{Q}\}$, R-charge $N_\Psi = \text{Tr } \Psi \bar{\Psi}$, $[N_\Psi, Q] = 3Q$ [Chen (1.1)–(1.2), (2.4)]. Hilbert space 2^{pN^2} -dimensional.

- **Pauli truncation.** Each mode appears at most once, so the algebra of words closes at finite length; at fixed N the constraint set is finite and the bootstrap can be *exact* at the top level. No flat directions, no continuum of moments.
- **Blocks.** M is block-diagonal in N_Ψ (as in LM §3.2.1: $F = 0, \pm 1, \pm 2$ blocks) and in $SU(N) \times \text{flavor}$ irrep. Fix a sector with $\phi(N_\Psi) = k$, $\phi(N_\Psi^2) = k^2$: since $\phi((N_\Psi - k)^2) = 0$, positivity forces $\phi((N_\Psi - k)O) = 0$ for every O in the set. Extremising $\phi(H)$ then bounds the sector ground energy $E_0(k)$ (*).
- **Which sector is the ground state.** For $Q = \text{Tr } \Psi^3$, $H = 3N(N^2 - 1) - 9\hat{C}_2$ [Chen (2.8)]: singlets are the *top* of the spectrum and the BPS states form the maximal-Casimir irrep r_* [Chen (2.18)–(2.22)]. A gauged (singlet) bootstrap bounds the wrong end. Work ungauged, with irrep-resolved blocks; the singlet-state justification for using open-index operators (LZ25 after (6)) must be replaced by the irrep-averaged functional ρ_R of KMPT (3.15).
- **SUSY floor and how to lift it** (*). With $Q, \bar{Q}$ in the basis, $\phi(H) \geq 0$ is free. In a lifted sector the SDP must additionally learn that annihilators act first on the sector ground state, e.g. $\phi(Q\bar{Q}) = 0$ at $N_\Psi = 0$. In a basis of elementary letters this follows from $\phi(N_\Psi) = 0$, $\phi(\Psi_{ij}\bar{\Psi}_{ji}) \geq 0$ and Cauchy–Schwarz. In a basis of *gauge-invariant single traces only* it does not: at $N = 3$, levels 3 and 4, all lifted sectors of the single-matrix model return the trivial bound 0 (verified 2026-09-15). Open-index (adjoint-valued) operators are required in the positivity matrix.
- **Ground-state positivity in a sector.** $\langle O^\dagger[H, O] \rangle \geq 0$ is valid for the sector ground state only if O preserves the sector ($[O, N_\Psi] = 0$, and likewise for the irrep); a charge-changing O may lower the energy by leaving the sector. For a genuine SUSY vacuum ($E = 0$) it is implied by the supercharge constraints (LZ24 App. A) and adds nothing.
- **Concentration as a bootstrap target** [CCSY Conj. 1, §2; Chen (2.22)]. R-charge concentration = one BPS degree per irreducible complex (labels: $N_\Psi \bmod 3$, $SU(N)$ irrep, flavor). A rigorous $E_0(k, \mathbf{r}) > 0$ outside a window of ≤ 3 consecutive k proves it; the single-matrix baseline has $E_0 = 0$ on $N + 1$ consecutive sectors.
- **Exact single-trace H for $Q = \text{Tr } \Psi^3$** (*, docs/derivations.md D1): $H = \frac{9}{2}(\text{Tr}[\Psi\Psi\bar{\Psi}\bar{\Psi}] + \text{Tr}[\bar{\Psi}\bar{\Psi}\Psi\Psi]) - \frac{3}{2}N(N^2 - 1)$, equivalently $\hat{C}_2 = NN_\Psi - \text{Tr } \Psi \text{Tr } \bar{\Psi} - \text{Tr}[\Psi\Psi\bar{\Psi}\bar{\Psi}]$. (An earlier fitted constant $-9(N - 1)^2$ is wrong for $N \geq 4$.)

6 Numbers to calibrate against

- Bosonic YM matrix QM (LZ25 Table II): $D=2$, $M=0$: $E \in [0.707832, 0.707868]$ at level 14; $D=9$: $E \in [6.69946, 6.69968]$, $\langle \text{tr } X_I X_I \rangle \in [2.29195, 2.29218]$ at level 11 (Monte Carlo 6.695(5), 2.291(1)). Islands first appear at level 10 for $M = 0$.
- BFSS (LZ24 Table 2): $\langle \text{tr } X^2 \rangle \geq 0.355$ at level 9 (Monte Carlo ≈ 0.378); analytic level 6: $\frac{3}{4}(3/50)^{1/3} \approx 0.2936$.
- Marinari–Parisi (LM §3.2.2): level 8, 44×44 , 27 free variables; $E_0/N^2 \approx 0.196 g^{2/3}$ (WKB 0.242); no upper bound obtained.
- Single-matrix fermionic model (Chen; this project): $E \in \{0, 18\}$ ($N=2$), $\{0, 18, 45, 72\}$ ($N=3$); $C_2^{\max} = N(N^2 - 1)/3$; $\dim r_* = 3^{N(N-1)/2}$; BPS window $[N(N-1)/2, N(N+1)/2]$. Three-matrix model at $N = 2$: 972 BPS states at $N_\Psi \in \{5, 6, 7\}$, counts 243:486:243, 168 distinct energies (ED, reproduced 2026-09-15).

References

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